LIMITS COMPUTED FROM GOOD DIAGRAMS OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Scheme-theoretic images. Morphisms, scheme-theoretic image lemmas

Source reference. Original reference 0CNM: sources/stacks-chow/tags.txt, source lines 15897–15897. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 0CNP: sources/stacks-chow/tags.txt, source lines 15899–15899. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01R8: sources/stacks-chow/tags.txt, source lines 2293–2293. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 056A: sources/stacks-chow/tags.txt, source lines 6703–6703. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0CNM. Let $X \to S$ be quasi-separated and of finite type. Given $i \in I$ and a diagram*

$$\begin{array}{ccc} X & \longrightarrow & W \\ \downarrow & & \downarrow \\ S & \longrightarrow & S_i \end{array}$$

as in (0CNP) for $i' \geq i$ let $X_{i'}$ be the scheme theoretic image of $X \to S_{i'} \times_{S_i} W$. Then $X = \lim_{i' \geq i} X_{i'}$.

Proof. Since X is quasi-compact and quasi-separated formation of the scheme theoretic image of $X \to S_{i'} \times_{S_i} W$ commutes with restriction to open subschemes (Morphisms, Lemma 01R8). Hence we may and do assume W is affine and maps into an affine open U_i of S_i. Let $U \subset S$, $U_{i'} \subset S_{i'}$ be the inverse image of U_i. Then U, $U_{i'}$, $S_{i'} \times_{S_i} W = U_{i'} \times_{U_i} W$, and $S \times_{S_i} W = U \times_{U_i} W$ are all affine. This implies X is affine because $X \to S \times_{S_i} W$ is a closed immersion. This also shows the ring map

$$\mathcal{O}(U) \otimes_{\mathcal{O}(U_i)} \mathcal{O}(W) \to \mathcal{O}(X)$$

is surjective. Let I be the kernel. Then we see that $X_{i'}$ is the spectrum of the ring

$$\mathcal{O}(X_{i'}) = \mathcal{O}(U_{i'}) \otimes_{\mathcal{O}(U_i)} \mathcal{O}(W)/I_{i'}$$

where $I_{i'}$ is the inverse image of the ideal I (see Morphisms, Example 056A). Since $\mathcal{O}(U) = \operatorname{colim} \mathcal{O}(U_{i'})$ we see that $I = \operatorname{colim} I_{i'}$ and we conclude that $\operatorname{colim} \mathcal{O}(X_{i'}) = \mathcal{O}(X)$. □

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